

Chapter 8

I/O Connector

Modern PC uses input and output ports to connect external devices to motherboard.

The most common input output ports are:

- Serial port
- Parallel port
- USB (Universal Serial Bus)

Serial Port

Serial ports transfer data one bit at a time. Data communication is done serially, not parallel. Each character that is sent over a serial connection is framed by start-and-stop signal. This way of communication is called asynchronous because synchronization between the two communicating devices is not needed. Serial ports could use synchronous way of communication as well.

What devices are connected to serial port? A lot of devices could be connected to serial port:

- modem
- barcode reader
- plotter
- mouse
- keyboard
- older type of printers
- computer

In serial ports, the communication is two-ways: from pc to the device and from device to PC.

There are different types of serial ports:

- Enhanced serial ports(ESP)
- Super high speed serial ports(SHSSP)

Most motherboards have at least one serial port. Serial port 1 is called COM1. If there are more than one serial ports, they are named as COM1, COM2, etc.

Serial Port Configuration

When you install devices on ports, you do not have to set IRQ, I/O address and DMA channel. This has been set already for ports during manufacturing. Serial ports use the following resources:

<u>COM X</u>	<u>I/O Ports</u>	<u>IRQ</u>
COM1	3F8-3FFh	IRQ4
COM2	2F8-2FFh	IRQ3
COM3	3E8-3EFh	IRQ4
COM4	2E8-2EFh	IRQ3

Types of Serial Ports

There are many serial port connectors:

- DB9
- DB15
- DB25

Parallel Port

Parallel ports are used to connect printers to PC in early days. Now parallel ports have become much more than that and used as more general purpose, relatively high speed interface between devices. Data is sent/transmitted eight lines at a time (a byte at a time). The data transfer rate is high because of parallel transmission. Parallel port transfer data only in one direction in early days i.e. from computer to the device though this is not the case today.

Some of the devices that could be connected include:

- Printer
- CD-ROM drives
- ZIP drives
- Scanner

Remark: Parallel port is used to connect printers. But it is not used exclusively for printers. Other devices can also be connected to parallel port as listed above.

Parallel ports are named as LPT1, LPT2 which means Line Printer Terminal. There are different types of parallel ports:

- unidirectional(standard parallel port)
- bidirectional
- Enhanced parallel port (EPP)
- Type 1(standard)
- Enhanced Capabilities Port (ECP)

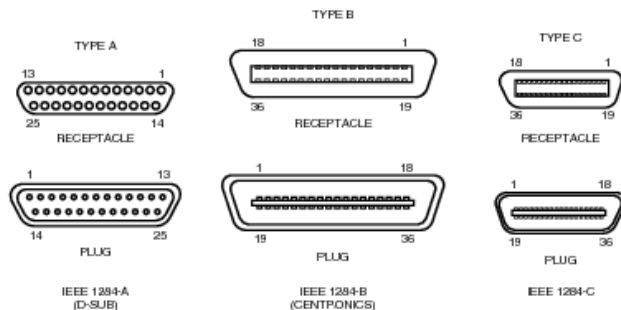


Fig parallel port

Unidirectional

In unidirectional parallel ports, data is transmitted only in one direction-from computer to the device. The device do not send feedback to the computer. But it is sometimes necessary to get feedback from devices to the computer.

Bidirectional

IBM introduced bidirectional parallel port in 1987. This type of port allows true communication i.e. the port allows printers to send information to computer unlike unidirectional port. The printer tells the computer the following:

- out of paper
- paper jam
- out of ink, etc

EPP

The Enhanced Parallel Port outperforms the standard parallel port. But it is compatible with standard port. Originally EPP is used primarily by peripherals other than printers like CD-ROM, hard disk drives, network adapters because they require constant two way communication with the PC.

- It offers high speed two way data transfers
- It handles hand shaking and synchronization between devices

ECP

-Like EPP, ECP offers improved performance for parallel port

- released in 1992

-developed by Hewlett Packard and Microsoft

-It is designed for high performance parallel communication for operation that involve moving large chunks of data but no monitoring like scanners, CD-ROM drives.

There are different types of parallel connectors:

-DB 25

-DB36

The ports have male and female specification. Male connectors have pins most of the time. The part of cable that fits into computer side port is male. The part of computer where we insert cable is female port. The male is inserted into female and connection is created.

DB25: It has 25 pins on the cable side and a place where it fits on the motherboard.

DB36:

Parallel Port configuration

Configuration is not needed for parallel port devices because this has been done during manufacturing. The resources used by parallel port devices:

<u>Standard</u>	<u>I/O port</u>	<u>IRQ</u>
LPT1	378-37Ah	IRQ7
LPT2	278-27Ah	IRQ5

USB

Universal Serial Bus is an external bus standard designed to bring plug and play to attach external devices to PC. It can help to connect external devices without the need to configure the device by installing on expansion card. This makes the configuration of new devices easier.



Fig USB logo

USB allows upto 127 devices to run simultaneously (at the same time). Intel has been the primary proponent of USB. USB transfers data both synchronously and asynchronously:

- 1.5MPS for devices such as mouse and keyboard
- 12MPS transfer rate for high-bandwidth devices such as modem, speaker, scanner, and monitors. You can connect USB devices while the computer is running while unlike other ports.

Input Devices

Keyboard and mouse are the most commonly used type of input device. Modern mouse and keyboard use USB port which is easy to use for external devices. But before the introduction of USB, they use their own port or serial port.

Keyboard

There are different layout of character on the keyboard. Based on this, keyboards are divided into:

- 101 key enhanced Keyboard
- 104 key windows keyboard
- 83 key PC and XT keyboard(obsolete)
- 84 key AT keyboard(obsolete)

The connectors used for keyboard can be

- 5 pin DIN connector
- 6 pin Mini-DIN connector
- USB
- COM1/COM2

PS/2 Connectors: a dedicated keyboard port.

Mouse

Mouse was invented by Douglas Engelbert in 1964. Since then it passed through many modifications. Today, mouse is one of the most commonly used input device. Mouse can be connected to the motherboard through one of the following:

- serial interface
- dedicated motherboard port(PS/2)
- Bus card interface
- USB port

Serial Interface:

Older PCs use serial port to connect mouse to the motherboard. The connector on the other end of cable is either 9 pin or 15 pin male connector. The mouse could be connected in COM1 or COM2.

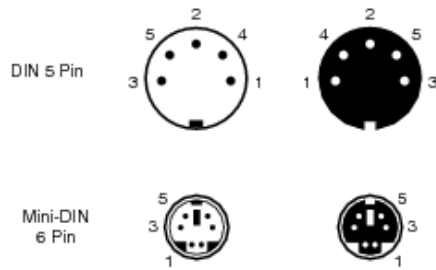


Fig Ps/2 Mouse ports

Dedicated Mouse Port(PS/2):

Most computer today come with a dedicated mouse port built into the motherboard. This dedicated interface is called PS/2mouse interface. It was introduced in 1987 by IBM.

Bus Card Interface:

Assume your mouse port has malfunctioned and you do not have any free COM port. What do you do? In such cases, you can install a mouse interface card on expansion slot and connect your mouse to the card.

Disadvantage:

- it occupies a slot that could have been used for other peripherals
- difficult to find such mouse or card in the market